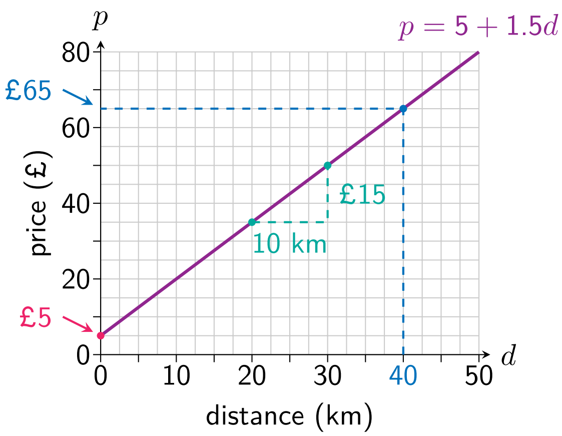
1.3 Linear Relationships

Some types of relationship are commonly encountered, and the rest of this chapter will study these in more detail. One of the simplest relationships between two variables is a *linear relationship*, which on a graph is a **straight line**. A linear relationship is defined by a **constant rate of change**, which can be calculated as the *gradient* of the graph. This tells us the amount of change in the response variable for each positive unit change in the independent variable.

For example, consider the linear relationship between the price of a taxi ride (£p) and the distance travelled (d km).



The **rate of change** is $\frac{15}{10} = £1.50$ per km, so each extra kilometre increases the price by £1.50.

The graph's **starting value**, for a distance of 0 kilometres is £5. (The taxi company's 'base charge').

Note that both the *starting value* and the *rate of change* can be observed from the relationship's **equation**:

$$\boxed{\textbf{price} = 5 + 1.5 \times \textbf{distance}}$$

For any equation of the form $y = a + bx$ describing a linear relationship between x and y, the value of a tells us the *starting value* of y (when $x = 0$), and the value of b tells us how much y changes for each 'step' in the x direction.

The *dashed line* shows how the graph can be used to **predict** a price of £65 for a distance of 40 kilometres.

The equation can also be used:

$$\text{price} = 5 + 1.5 \times 40 = 65$$

Example

Problem:
A candle with a height of 20 centimetres is lit. It loses 4 centimetres of height per hour as it burns.

(a) Write down an equation connecting the candle's height and its burn time.

(b) State the units used to measure the rate of change.

(c) State which is the independent variable and which is the response variable.

(d) Use the equation to calculate the height of the candle after 90 **minutes** spent burning.

(e) Draw a graph to show the relationship.

Solution:

(a) height $=$ 20 $-$ 4 $\times$ time

(b) Centimetres per hour.

(c) The independent variable is time.
The response variable is height.

(d) height $=$ 20 $-$ 4 $\times$ 1.5 $=$ 14 cm

(e)